

Financial Time Series

46-929

Solutions #2

1. Weak Stationarity

- a) (Differencing preserves weak stationarity)

Suppose $\{X_t, -\infty < t < \infty\}$ is a weakly stationary process. Define a new process $Y_t = (1 - B)X_t = X_t - X_{t-1}$. Show that $\{Y_t, -\infty < t < \infty\}$ is also weakly stationary and find the mean, variance and autocovariance function of the Y_t process.

- b) (Linear combinations of independent weakly stationary processes are weakly stationary)

Suppose $\{X_t^{(i)}, -\infty < t < \infty, i = 1, 2\}$ are weakly stationary independent. Show that $\{Y_t, -\infty < t < \infty\}$ where $Y_t = aX_t^{(1)} + bX_t^{(2)} + c$ is also weakly stationary.

Solution

- a) Suppose X_t process is weakly stationary with autocovariance function $\gamma_X(h)$ and $Y_t = X_t - X_{t-1}$. Then

$$\begin{aligned} \text{Cov}(Y_t, Y_{t+h}) &= \text{Cov}(X_t - X_{t-1}, X_{t+h} - X_{t-1+h}) \\ &= \text{Cov}(X_t, X_{t+h}) - \text{Cov}(X_t, X_{t-1+h}) - \text{Cov}(X_{t-1}, X_{t+h}) + \text{Cov}(X_{t-1}, X_{t-1+h}) \\ &= \gamma_X(h) - \gamma_X(h-1) - \gamma_X(h+1) + \gamma_X(h) \end{aligned}$$

Thus $\gamma_Y(h) = 2\gamma_X(0) - \gamma_X(h-1) - \gamma_X(h+1)$ which is independent of t . Also the mean $E(Y_t) = 0$, so the mean and variance are constant and the covariance function depends only upon the time lag. Hence Y_t is weakly stationary.

- b) The calculation is similar to part a). Because $\{X_i, i = 1, 2\}$ are weakly stationary, they have constant mean μ_i , constant variance σ_i^2 and are independent. Consequently, $E(Y_t) = a\mu_1 + b\mu_2 + c$ and $\text{Var}(Y_t) = a^2\sigma_1^2 + b^2\sigma_2^2$, both independent of t . Since the constant mean of Y_t and the constant c will not impact the autocovariance function of the Y process, we compute

$$\begin{aligned} \text{Cov}(Y_t, Y_{t+h}) &= \text{Cov}(aX_t^{(1)} + bX_t^{(2)}, aX_{t+h}^{(1)} + bX_{t+h}^{(2)}) \\ &= a^2\text{Cov}(X_t^{(1)}, X_{t+h}^{(1)}) + b^2\text{Cov}(X_t^{(2)}, X_{t+h}^{(2)}) + 2ab(0) \\ &= a^2\gamma_{X^{(1)}}(h) + b^2\gamma_{X^{(2)}}(h). \end{aligned}$$

Thus the autocorrelation function of $\{Y_t\}$ is independent of t . Consequently, the process is weakly stationary.

2. Introduction to the Yule Walker Equations

In lecture #3, we found that for an AR(2) process, $\{X_t, -\infty < t < \infty\}$ with $\{\epsilon_t, -\infty < t < \infty\}$ a $\text{WN}(0, \sigma_\epsilon)$ process, there was linear relationship between the model coefficients ϕ_1, ϕ_2 and the autocovariance function $\gamma(h)$ recalling that $\gamma(0) = \sigma_\epsilon^2$ and $\gamma(-h) = \gamma(h)$. This was used to find an iteration by which one could express $\gamma(h)$ or $\rho(h)$ as a function of ϕ_1 and ϕ_2 for any h .

- a) Turning this around, find two linear equations for ϕ_1, ϕ_2 in terms of ρ_1 and ρ_2 . Express those two equations in matrix form. Solve to find a matrix expression for ϕ_1 and ϕ_2 .
- b) Suppose we have estimates of $\gamma(1)$ and $\gamma(2)$, namely $\hat{\gamma}(1)$ and $\hat{\gamma}(2)$. One could plug them into your expression from part a) to obtain estimates for ϕ_1 and ϕ_2 . Simulate 1,000 data points from an AR(2) sequence with iid $N(0, \sigma^2)$ errors and using those data, estimate ϕ_1 and ϕ_2 . Use

$\phi_0 = 0, \phi_1 = .33, \phi_2 = .33, \sigma_\epsilon^2 = .5$. Note, you can use the Python routine `statsmodels.regression.linear_models.yule_walker(data, order=2, method='mle')` as this 2x2 system is a very simple version of the general Yule-Walker equations to be covered in class.

- c) The accuracy of this method (assuming the data actually conforms to the model) depends upon σ_ϵ^2 and n , the size of the data. Repeat part b) with $n = 100$ and $n = 500$ to get an idea of how the error of estimation changes with the sample size.

Solution

- a) It is convenient to remove any mean term as it will not change the autocorrelation of the process, so we set $Y_t = X_t - \mu$, where $E(X_t) = \mu$. $\rho_X(i) = \rho_Y(i)$. As was presented in lecture, we start with the AR(2) model,

$$y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t.$$

Multiplying across by first Y_{t-1} then Y_{t-2} , taking expectations and dividing by $\gamma(0)$, we find

$$\rho(1) = \phi_1 \rho(0) + \phi_2 \rho(1) + 0,$$

$$\rho(2) = \phi_1 \rho(1) + \phi_2 \rho(0) + 0,$$

since Y_s and ϵ_t are uncorrelated for $s < t$. Recalling that $\rho(0) = 1$, these two equations can be put into matrix form with $\boldsymbol{\rho} = (\rho_1, \rho_2)^T$, $\boldsymbol{\phi} = (\phi_1, \phi_2)^T$, and

$$\mathbf{R} = \begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix},$$

$$\boldsymbol{\rho} = \mathbf{R}\boldsymbol{\phi}$$

This can be solved by

$$\boldsymbol{\phi} = \mathbf{R}^{-1}\boldsymbol{\rho}.$$

- b)c) In a single run, the results were $\hat{\phi}_1 = .3036$ and $\hat{\phi}_2 = .3391$. One would need to run this routine m times to get some idea of the variability of the estimate. Below in a table are the estimates for $m = 10$ runs using $\sigma = .5$ for $n = 100, 500, 1000$. The results are the mean over 10 runs for each of the 3 cases. The estimates are given along with the estimated standard deviations (the sample standard deviation, not the standard error of the parameter estimates, which would need to be divided by $\sqrt{n}$).

n	100	500	1000
$\hat{\phi}_1$	.3544 (.1092)	.3518 (.0324)	.3370 (.0305)
$\hat{\phi}_2$	.2840 (.0849)	.3150 (.0434)	.3055 (.0280)

3. MA(1)

Consider a MA(1) process with mean 0,

$$X_t = \theta_1 \epsilon_{t-1} + \epsilon_t,$$

for ϵ_t a $WN(0, \sigma^2)$. Pick your own coefficient θ_1 and let $n = 1000$. We worked out the autocovariance and autocorrelation function for some specific cases including MA(1). If one has a moment estimator for $\gamma(1)$, then one can use this to find a moment estimator for θ_1 . Note, this can be done for higher-order MA processes, but the resulting equations are non-linear and difficult to solve. Simulate the MA(1) series of (length 1,000) 10 times with $\sigma^2 = 1$ each time using the simulated data to estimate θ . Report your 10 estimates and the variability of those estimates.

Solution

The MA(1) process is given by the equation

$$X_t = \mu + \theta_1 \epsilon_{t-1} + \epsilon_t.$$

As before, we set $\mu = 0$ since it will not change the autocovariance function. We multiply both sides by X_{t-1} and take expectations. This gives $E(X_t X_{t-1}) = \theta_1 E(X_{t-1} \epsilon_{t-1}) + E(\epsilon_t X_{t-1})$. $E(\epsilon_t X_{t-1}) = 0$, because X_{t-1} is a linear function of the ϵ_s , $s < t$, so X_{t-1} and ϵ_t are uncorrelated. Thus $\gamma(1) = \theta_1 \sigma_\epsilon^2$. Dividing by $\gamma(0) = (1 + \theta_1^2) \sigma_\epsilon^2$ we find

$$\rho(1) = \frac{\theta_1}{1 + \theta_1^2}$$

. This gives a quadratic equation for θ_1 as a function of $\rho(1)$. If we have a moment estimate of $\rho(1)$, then we can plug it into the solution of the quadratic to get an estimate of θ_1 . Specifically,

$$\hat{\theta}_1 = \frac{1 - \sqrt{1 - 4r^2}}{2r},$$

where r is the moment estimate of $\rho(1)$. We pick the root so that $|\theta_1| < 1$, so the process is invertible.

The results from 10 simulations of θ_1 using this method with $\theta_1 = .25$ and $\sigma_\epsilon = 1$ had a mean value of .2660 and a standard deviation of .0249 with a range of [.2395,.3109].

4. Spurious Regression

The problem of spurious regression (and spurious correlation) arises especially in economics where one performs a regression analysis with non-stationary time series data, and most economic time series are non-stationary. The simplest situation is where one has two non-stationary time series that are independent and regresses one on the other. Because the two series, while being independent, are both related to time, a significant but spurious correlation or regression can be found. Time is a confounding factor. The first report of this phenomenon dates back to G. Udny Yule in 1926, “Why do we Sometimes get Nonsense-Correlation between Time Series? – A Study in Sampling and the Nature of Time-Series.” The most cited paper on this topic is the 1974 paper by Granger and Newbold, “Spurious Regressions in Econometrics,” (both papers are in the Course Papers folder). The simplest version of this phenomenon is where $\{X_t\}$ and $\{Y_t\}$ are independent random walks (which are non-stationary). That is $X_t = \sum_{i=1}^t Z_i^{(1)}$ and $Y_t = \sum_{i=1}^t Z_i^{(2)}$ where $\{(Z_i^{(1)}, Z_i^{(2)})\}$ are iid with independent $N(0, \sigma_j^2)$, $j = 1, 2$ distributions. Consider the model

$$Y_i = \beta_0 + \beta_1 X_i, \quad 1 \leq i \leq t.$$

Clearly because the X and Y sequences are independent, $\beta_1 = 0$; however, one can still test $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$. Granger and Newbold did this 100 times with simulated data and found that roughly 75% of the trials resulted in rejection of H_0 . In this problem let $\sigma_i^2 = 1$, $i = 1, 2$, simulate $\{(X_t, Y_t), 1 \leq t \leq 50\}$ and test whether H_0 is significant at the 5% level. Do this 100 times and report the results.

Solution

Granger and Newbold fit a simple linear regression model $Y = \beta_0 + \beta_1 X + \epsilon$ with a sample size of 50 for two independent random walks for X and Y . They tested $H_0 : \beta_1 = 0$ at the .05 α level. They reported that “three-quarters” of the 100 trials resulted in rejection, indicating a spurious regression. While the problem called for replication of the results, the results below are for a larger sample size ($n = 10,000$), show both $\alpha = .05$ and .01 and use the size of the data to be 50, 100 and 250. One can see that the “three-quarters” probability of a spurious result cited by Granger and Newbold is too high (presumably due to sampling variability) as “two-thirds” is closer. Also increasing the sample size leads to more frequent spurious regression. Also reducing the α level from .05 to .01 reduces the frequency of spurious regression; however, it is still a very frequent phenomenon. The python program below will produce the results in the table.

α	$n = 50$	$n = 100$	$n = 250$
.05	.667 (.005)	.761 (.004)	.847 (.004)
.01	.574 (.005)	.689 (.005)	.803 (.004)

```

import numpy as np
import statsmodels.api as sm

n=10000
reject = 0
for i in range(n):
    z1 = np.random.randn(50)
    z2 = np.random.randn(50)
    x = np.cumsum(z1)
    y = np.cumsum(z2)
    x_one = sm.add_constant(x)
    model = sm.OLS(y,x_one).fit()
    if model.pvalues[1] < 0.05:
        reject = reject+1

print(rej)
p = rej/n
print(np.sqrt(p*(1-p)/n))

```